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Temperature-Sensitive Juvenile Fitness Traits Show Correlated and Genetically Defined Variation in a Heatwave-Affected Marine Fish

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ABSTRACT

Juvenile fish must balance energy allocation between somatic growth and lipid storage. This trade-off is sensitive to environmental conditions, which can drive interannual differences in overwinter survival and recruitment. We examined individual variation in these strategies by rearing Gulf of Alaska (GOA) juvenile Pacific cod (*Gadus macrocephalus*) at four temperatures (0°C, 5°C, 9°C, and 16°C). Growth rates followed an asymmetric thermal performance curve with a peak near 12°C, while liver lipid content, the energy currency of juvenile cod, declined sharply above 9°C. Individuals at 16°C stored on average approximately 40% less lipid in the liver than those held at cooler temperatures, despite high survival (90%). However, within each temperature, individuals with the highest growth rates also had the greatest liver lipid reserves, suggesting variation in energetic efficiency among individuals. At the highest temperature (16°C), ~4% of genome-wide genetic variation captured by three genetic principal components explained ~25% of trait variation, with particularly strong relationships to liver lipid content. These results highlight that intra-population genetic variability associated with individual fitness traits, particularly lipid allocation in the liver, may influence resilience to warming in GOA juveniles. As marine heatwaves become more frequent, reduced lipid stores in the liver may compromise overwinter survival, while traits supporting both growth and energy storage may be favored. This study provides insight into thermal lipid allocation strategies in juvenile Pacific cod, and highlights the potential for developing molecular indicators of selection on fitness-related traits as marine conditions continue to shift.

1 | Introduction

In fish, high temperature accelerates early life growth and development but leads to trade-offs later in life by limiting adult body size, maximum size, and reproductive output (Huss et al. 2019; Atkinson 1994). These patterns are driven by thermal regulation of metabolic processes, which determine energetic resources after accounting for maintenance costs—costs that increase disproportionately with temperature and body size (Huey and Kingsolver 2019; Ohlberger et al. 2011; Christensen et al. 2020; Lindmark et al. 2022). While much attention has been given to energy trade-offs between growth and reproduction, less is

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known about how temperature mediates allocation trade-offs between somatic growth and energy storage (e.g., lipids in the liver) in early life stages.

Trade-offs between growth and lipid storage as fish prepare for winter have been explored in some systems (Biro et al. 2005), but there is evidence that such trade-offs are highly variable and may be species- and season-specific, particularly in relation to shifting thermal environments (Copeman et al. 2017; Abookire et al. 2024). Empirical studies have taken a population-level approach which overlooks temperature-driven physiology and phenotypic responses at the individual level (Pörtner and Farrell 2008; Peck et al. 2013). Small differences in size and condition during early life stages can strongly influence survival (Sogard 1997; Houde 1987), and energetic tradeoffs during juvenile development are likely to shape maturity schedules and lifetime reproductive output (Ivan and Höök 2015; Hutchings 2002). If high temperatures favor certain energy allocation strategies (e.g., faster growth vs. increased lipid reserves), this would represent a key mechanism by which environmental variability affects a population through selection on juvenile traits and their underlying genetic factors.

In the Gulf of Alaska (GOA) a series of marine heatwaves (MHWs) in 2014–2016 and again in 2019 triggered widespread ecological disruptions (Suryan et al. 2021), including severe population declines in Pacific cod (*Gadus macrocephalus*). These events led to failed recruitment, dramatic losses in adult biomass, and ultimately the closure of the federal Pacific cod fishery (Barbeaux et al. 2020; Laurel and Rogers 2020). However, in the years following these MHWs, signs of strong Pacific cod recruitment have emerged in nursery habitats (Laurel et al. 2023; Kalei Shotwell and Dame 2023), along with increases in average juvenile size (Almeida et al. 2024) and the discovery of previously undetected genetic structure, potentially signaling resilience within the population (Laurel, Miller, et al. 2026). Despite these encouraging signs, it remains unclear whether these post-MHW cohorts reflect selection for individuals better adapted to thermal stress, or whether they may be more vulnerable during overwintering due to trade-offs between growth and lipid storage. For instance, if faster-growing age-0 cod grow at the expense of lipid reserves, they may have lower survival probabilities during warm winters than slower-growing juveniles in better energetic condition (Laurel, Hicks, et al. 2026).

The goal of this study was to examine how temperature shapes energy allocation strategies in juvenile Pacific cod, with particular focus on responses to high temperature. We address three related questions using individual-level measurements of growth, liver lipid storage, and genomic variation. First, how does temperature influence growth and lipid storage in juvenile Pacific cod? Previous laboratory studies conducted prior to the MHWs established baseline relationships between temperature, growth, and lipid accumulation in juvenile Pacific cod using tank-level averages (Laurel, Spencer et al. 2016; Copeman et al. 2017); here we re-establish these models using individual-level data, and with a post-heatwave era cohort. These earlier findings also provide a point of comparison for assessing whether post-MHW cohorts exhibit similar or shifted energetic responses to thermal gradients. Second, how are growth and energy storage related within and among individuals across a thermal gradient? Characterizing these relationships allows us to determine how (or if) energetic trade-offs vary by temperature. Third, to what extent is variation in fitness traits associated with underlying genetic variation?

Addressing these questions could allow us to assess whether shifts or variation in temperature sensitivities have altered, and will continue to alter, the survival trajectories of juveniles, and whether they could play a key role in shaping the future stability of GOA Pacific cod populations.

2 | Methods

2.1 | Temperature Treatments and Tissue Sampling

Live juvenile Pacific cod (age-0) were collected off Kodiak Island, AK on August 25, 2022 (Figure 1) and transported to the NOAA laboratory in Newport, OR. Fish were acclimated for 3 months in a 7500-L flow-through seawater tank at 6°C. On November 21, 2022, individuals were implanted with 8-mm passive integrated transponder (PIT) tags (BioMark) under MS-222 anesthesia (50 ppm, 60 s). Following recovery, fish were returned to their rearing tanks. Tag retention was 100% and no tagging-related mortalities occurred prior to initiation of the experiment 5 weeks later. Standard length and wet weight were measured at three time points: prior to acclimation, after acclimation (before temperature adjustment), and at treatment termination. Mortality was monitored daily.

On December 27, 2022, juveniles ($N = 160$, 89.3 ± 7.3 mm standard length; 8.1 ± 1.8 g wet weight) were distributed among sixteen 50-L tanks assigned to 0°C, 5°C, 9°C, or 16°C (40 fish per temperature distributed across four tanks). Temperatures were chosen to replicate those used in previous studies (Copeman et al. 2017; Almeida et al. 2024), which broadly encompass the thermal regime across the Pacific cod range. Following a 1-week acclimation at 6°C, temperatures were adjusted by ≤ 1°C per day until targets were reached (January 8, 2023), after which they were maintained for 4 weeks. Photoperiod was maintained at 12:12 h, and fish were fed to satiation five times weekly with shredded gel food and krill.

On February 8–9, 2023, surviving fish were euthanized with MS-222 and immediately placed on ice to minimize lipid degradation. Fish were rinsed and measured, then eviscerated and livers excised and weighed to calculate hepatosomatic index (HSI). Approximately 100 mg of liver tissue was subsampled from both lobes, placed in solvent-cleaned 7 mL glass vials, submerged in 2 mL chloroform under nitrogen, and stored at −20°C until lipid analyses. Fin clips were preserved in 100% non-denatured ethanol for genetic analyses. Measurement resolution was ±0.5 mm (length), ±0.01 g (body mass), and ±0.0001 g (liver mass).

2.2 | Genetic Composition

Experimental fish were wild-caught near Kodiak Island, Alaska, a region that can contain juvenile offspring from a mixture of spawning stocks (Schaal et al. 2026). To assess population of

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origin and identify variation potentially associated with thermal response, we performed low-coverage whole-genome sequencing (WGS) on all experimental individuals ($n = 160$), and compared these data to a previously generated low-coverage WGS dataset of adult reference fish ($n = 633$) from four major Alaskan marine regions (Schaal et al. 2026). Full sequencing and bioinformatic methods are provided in Appendix S1. Briefly, genomic DNA was extracted from fin clips using the DNeasy 96 Blood & Tissue Kit (Qiagen Inc., Valencia, CA) and sequenced across four lanes of an Illumina NovaSeq 6000 platform to generate 150-bp paired-end reads. Raw reads were trimmed, filtered, and aligned to the Pacific cod reference genome (ASM3116895v1), resulting in a mean depth of ~3× per sample.

To assess the likely population of origin, genotype likelihoods were calculated jointly across experimental and reference samples using ANGSD v0.940 (Korneliussen et al. 2014). Population was assessed using principal component analysis (PCA) based on a covariance matrix estimated with PCAngsd v1.36.4 (Figure 2).

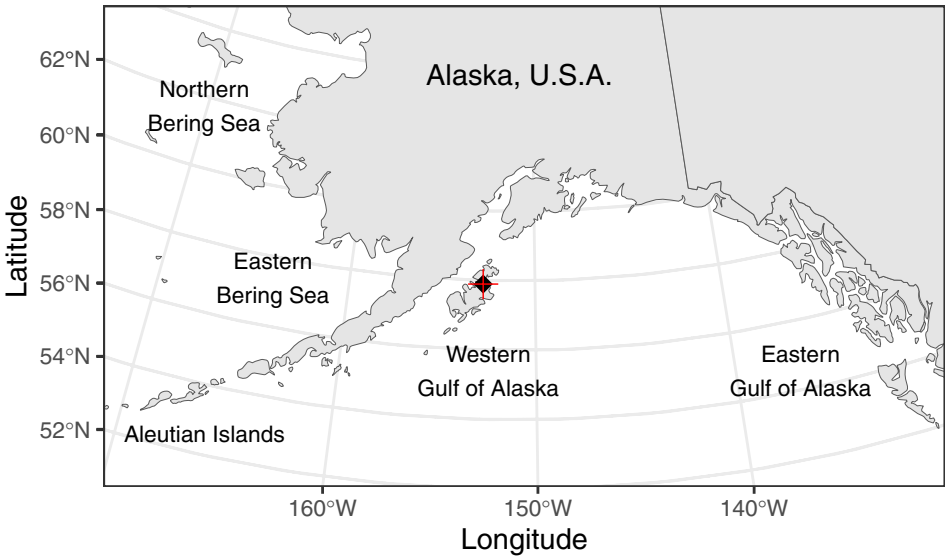


FIGURE 1 | Location of origin for experimental Pacific cod juveniles (age-0), which were collected from nursery grounds by beach seine off Kodiak Island, AK (black/red point) in August 2022.

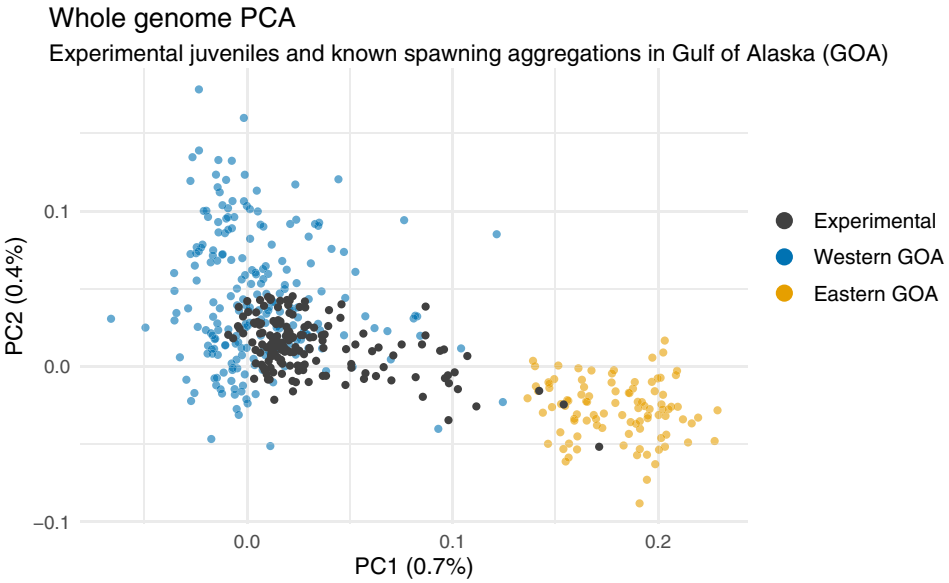


FIGURE 2 | Principal component analysis (PCA) biplot constructed from genotype likelihoods at all polymorphic sites ($n = 285{,}061$) shared among experimental juveniles and reference Pacific cod spawning aggregations. Of the 160 wild-caught experimental juveniles, 159 were genetically similar to the Gulf of Alaska (GOA) spawning aggregations, with the majority clustering with the western GOA group ($n = 128$), three with the eastern GOA group, and appr. twenty-eight intermediate between the two. One juvenile was excluded from all analyses because it was a genetic outlier, indicating likely origin from another population (not shown). Reference individuals from additional regions (Aleutian Islands and Bering Sea) were included in the initial PCA but omitted here to focus on the GOA populations relevant to the experimental fish. See Figure S1 for a PCA of experimental juveniles only, based on a larger set of sites that passed filtering when reference adults were excluded ($n = 1.6$ M).

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TABLE 1 | Sample sizes (number of individuals with data) per temperature treatment and per data set. Temperatures represent the mean ± 1 standard deviation (SD), averaged across all tanks over the duration of the treatment period.

Temperature treatment	0.0°C ± 0.1°C	5.0°C ± 0.2°C	9.1°C ± 0.2°C	15.9°C ± 0.2°C
Genetics (WGS, 3× depth)	40	39	40	40
Growth and condition indices	40	39	40	36
Liver lipids assays	24	24	25	25

Note: Each temperature treatment included 40 individuals distributed across four tanks ($n = 10$ per tank), for a total of 16 tanks and 160 individuals. One individual from 5°C was excluded from all analyses as it was a genetic outlier (likely not from the GOA populations). Four individuals in the 16°C treatment died prior to the end of the experiment and were excluded from growth, condition, and lipid analyses, as they were not comparable to individuals sampled at the end of the experiment. Lipid analyses were conducted on a subset of individuals from each treatment after these exclusions.

For genotype-trait analyses, genotype likelihoods were recalculated using experimental fish only ($n = 159$, one genetic outlier excluded) across all variable loci (i.e., not restricted to sites retained when including reference adults), and PCA was performed on this dataset. The resulting genetic PC axes were used as explanatory variables in redundancy analyses (see Section 2).

2.3 | Liver Lipid Extraction

Lipid mass fractions were measured in liver tissues for a subset of individuals from each treatment ($N = 98$, Table 1). Liver tissue was used because hepatic lipid content is highly responsive to temperature in juvenile Pacific cod, whereas muscle lipid mass fraction varies comparatively little under similar thermal treatments (Copeman et al. 2017, 2013, 2008). Livers were stored at −20°C and processed within 6 months of sampling. Total lipids were extracted using a modified Folch procedure following Parrish (Parrish 1987; Folch 1957). Lipid classes were quantified by thin-layer chromatography with flame ionization detection (TLC-FID) using a MARK VI Iatroscan (Iatron Laboratories) as described previously (Copeman et al. 2017; Lu et al. 2008). Extracts were applied to activated silica-coated Chromarods and separated using a multi-stage solvent development system to quantify triacylglycerols (TAG), free fatty acids (FFA), sterols (ST), and polar lipids (PL). Lipid classes were quantified using lipid-specific calibration curves. Commercial standards (Sigma-Aldrich) were used for FFA, ST, and PL, and a gadid-specific liver TAG standard was purified via column chromatography following Ohman (Ohman 1997). Detailed solvent compositions and development conditions are provided in Appendix S1.

2.4 | Growth, Condition, and Liver Lipid Data Analyses

Standard length, wet weight, Fulton's condition factor (K_{wet}), and hepatosomatic index (HSI) were quantified in 155 individuals (Table 1). One individual from the 5°C treatment was excluded as it was a genetic outlier, and four individuals from the 16°C treatment were excluded because they died prior to experiment completion.

Specific growth rates (SGR) were calculated as described in Crane et al. (2020) for each individual during both the acclimation and experimental periods. SGRs were based on standard length (SGR_{SL}) and wet weight (SGR_{WW}) using $SGR = 100 \times (e^{g} - 1)$ where $g = \frac{lnM_2 - lnM_1}{t_2 - t_1}$ and M_1 and M_2 are measurements taken before (t_1) and after (t_2) the acclimation period to assess baseline growth rates, and before and after experimental temperature treatments to assess effects of temperature. Fulton's condition index (K_{wet}) was approximated as $K_{wet} = 100 \times \frac{\text{wet total weight (g)}}{(\text{standard length (mm)}/10)^3}$. The hepato-somatic index (HSI) was calculated as $HSI = 100 \times \frac{\text{wet liver weight (g)}}{\text{wet total weight (g)}}$.

Liver lipid components were analyzed as mass fractions relative to liver tissue wet weight ($\mu g \cdot mg^{-1}$). Total liver lipid content was estimated by summing measured liver lipid classes (triacylglycerol, free fatty acids, sterols, and polar lipids), multiplying by total liver wet weight to estimate lipid mass per fish (mg).

To identify growth and liver metrics affected by temperature, we fit linear mixed-effects models with temperature as a categorical fixed effect and with tank included as a random intercept. Models were fit using restricted maximum likelihood (REML), and temperature effects were evaluated using ANOVA F-tests. Model assumptions were assessed using residual diagnostics. Polar lipid content exhibited right skew and was log-transformed prior to analysis; all other variables were analyzed on their original scale. Because smaller juveniles may exhibit faster growth at this life stage (Hurst et al. 2010), initial body size (scaled standard length) was included as a covariate in models of growth rate, HSI, K_{wet}, and total liver lipid content. We also explicitly tested for size-dependent growth by fitting additional mixed-effects models including initial body size (wet weight or standard length), temperature, and tank as a random intercept.

Temperature-dependent models were then developed for all metrics with significant temperature effects, using best-fit models limited to three parameters following Laurel et al. (2016). Individuals were used as the level of observation. Candidate models included linear, quadratic, and cubic polynomial regressions, as well as Gaussian thermal performance curves (following Copeman et al. 2017). Models were compared using likelihood ratio tests and Akaike's Information Criterion (AIC), and the most parsimonious model was selected. Goodness of fit was assessed using pseudo-R^2 values.

2.5 | Relationships Among Growth Traits, Lipid Traits, and Genetics

Pearson correlation coefficients were calculated among growth and lipid traits across all individuals and separately within each temperature treatment. p values were adjusted using false

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discovery rate (FDR) correction within each group to account for multiple testing.

Redundancy analysis (RDA) was conducted to evaluate relationships among genetic and phenotypic variation at the highest, most thermally stressful temperature (16°C). Individuals subjected to this temperature were included if they had complete data for seven physiological traits ($n = 25$): specific growth rates based on weight, hepatosomatic index (HSI), liver lipid content, and mass fractions of liver lipid classes (triacylglycerols, free fatty acids, sterols, and polar lipids). Trait variables were standardized (z-score transformed) prior to analysis. Based on the distribution of variance explained across axes, we limited the analysis to the top 10 PCs, which each explained unique structure in genetic composition and >1% of total variance, and cumulatively explained 12.1% of total genetic variance; PCs beyond PC10 explained uniform and minimal variance (<0.6%, Figure S2). Model selection was performed using backward stepwise procedures (*ordistep* function in the R package vegan) to identify the top candidate explanatory variables (PC axes), and the final model was selected with the highest adjusted R^2. Model significance was assessed using permutation tests (999 permutations). Pearson correlations were calculated between traits and genetic PCs retained in RDA models.

2.6 | Functional Characterization of Trait-Associated Genetic Variation

For principal components (PCs) that explained significant variation in physiological traits, we examined whether SNPs with the highest loadings occurred in genes associated with particular biological functions. SNP loadings from PCAngsd were standardized for each PC axis, and SNPs with extreme loadings ($|z| > 3$) were identified as candidates contributing strongly to each axis. Using genome coordinates for SNPs and annotated gene features from the NCBI genome annotation, SNPs located within gene bodies or within 2 kb upstream of genes (putative promoter regions) were assigned to genes. Genes were then filtered for those containing at least two SNPs.

To interpret gene function, Pacific cod gene sequences were compared to the UniProt/SwissProt database using BLAST (Lu et al. 2008), allowing assignment of UniProt accession numbers and associated functional annotations. Functional enrichment analysis was then performed using DAVID v6.8 (2021 update) (Sherman et al. 2022) to identify biological processes overrepresented among candidate genes relative to the full set of annotated Pacific cod genes. Gene Ontology (GO) biological processes and UniProt keywords were considered significantly enriched if they had an EASE score (modified Fisher's exact test) <0.05 and were represented by at least three genes.

3 | Results

3.1 | Genetic Composition

Principal component analysis was conducted with genotype likelihoods at all polymorphic sites ($n = 285{,}061$) shared among experimental juveniles and reference Pacific cod spawning aggregations. PCA clustering indicated that, of the 160 experimental juveniles, 159 were genetically similar to Gulf of Alaska (GOA) spawning aggregations (Figure 2). The majority of experimental juveniles ($n = 128$) clustered with the western GOA population, three clustered with the eastern GOA population (one in the 0°C treatment and two in the 9°C treatment), and 28 were intermediate between the two. Among these intermediate individuals, 9, 5, 7, and 7 were exposed to the 0°C, 5°C, 9°C, and 16°C treatments, respectively. One juvenile was excluded from analyses because it was a genetic outlier, indicating likely origin from another population.

A separate PCA was conducted using genotype likelihoods at all polymorphic sites identified in experimental individuals only ($n = 1.6$ M, Figure S1). PC1 in the new projection captured the same genetic structure as PC1 in the PCA conducted with reference adults (i.e., reflecting variation among wGOA and eGOA cod, with some of mixed ancestry). No genetic differences were detected among temperature treatments (Figure S1).

3.2 | Survival, Growth, Condition, and Liver Lipid Content

All juveniles survived the 0°C, 5°C, and 9°C treatments. In the 16°C treatment, four mortalities occurred on days 14, 16, 21, and 24 after target treatments were reached from three different tanks, resulting in a survival rate of 90%. Growth rates did not differ among temperatures during the 1-month acclimation period (between tagging and beginning temperature adjustments) and were not significantly associated with initial body size (wet weight: $F_{3,12.1} = 0.72$, $p = 0.56$; standard length: $F_{13,12.1} = 0.44$, $p = 0.73$).

During the experimental treatments, specific growth rates (SGR) based on standard length (SL) and wet weight (WW) differed significantly among temperatures (SL: $F_{3,12.8} = 82.6$, $p < 0.001$; WW: $F_{3,12.7} = 78.6$, $p < 0.001$). Solving the first derivative of the temperature-dependent growth models predicted maximum growth rates at 12.24°C and 12.18°C for weight- and length-based measurements, respectively (Figure 3, Table 2). Fulton's condition factor based on wet weight (K_{wet}) did not differ significantly among temperature treatments ($F_{3,12.3} = 1.1$, $p = 0.40$).

Hepatosomatic index (HSI), a measure of liver size relative to total body size (using wet weight), decreased with increasing temperature and differed significantly among temperatures ($F_{3,12.1} = 21.6$, $p < 0.001$). Total liver lipid mass fraction (μg/mg WWT) differed significantly among temperatures ($F_{3,11.6} = 7.7$, $p = 0.004$), with the highest values observed at 9°C and lowest at 16°C. This pattern was largely driven by variation in triacylglycerol content ($F_{3,11.8} = 7.3$, $p = 0.005$). Sterols ($F_{3,11.8} = 4.98$, $p = 0.018$) and free fatty acids ($F_{3,93} = 3.5$, $p = 0.050$) also varied among treatments, but each comprised only a minor fraction of total liver lipids (~1%–2%; Table 2). Temperature-dependent models predicted maximal lipid mass fractions at 6.6°C for triacylglycerols, 6.5°C for free fatty acids, and 4.5°C for sterols (Figure 3, Table 2). Polar lipids did not vary significantly with temperature ($F_{3,12.9} = 2.6$, $p = 0.10$).

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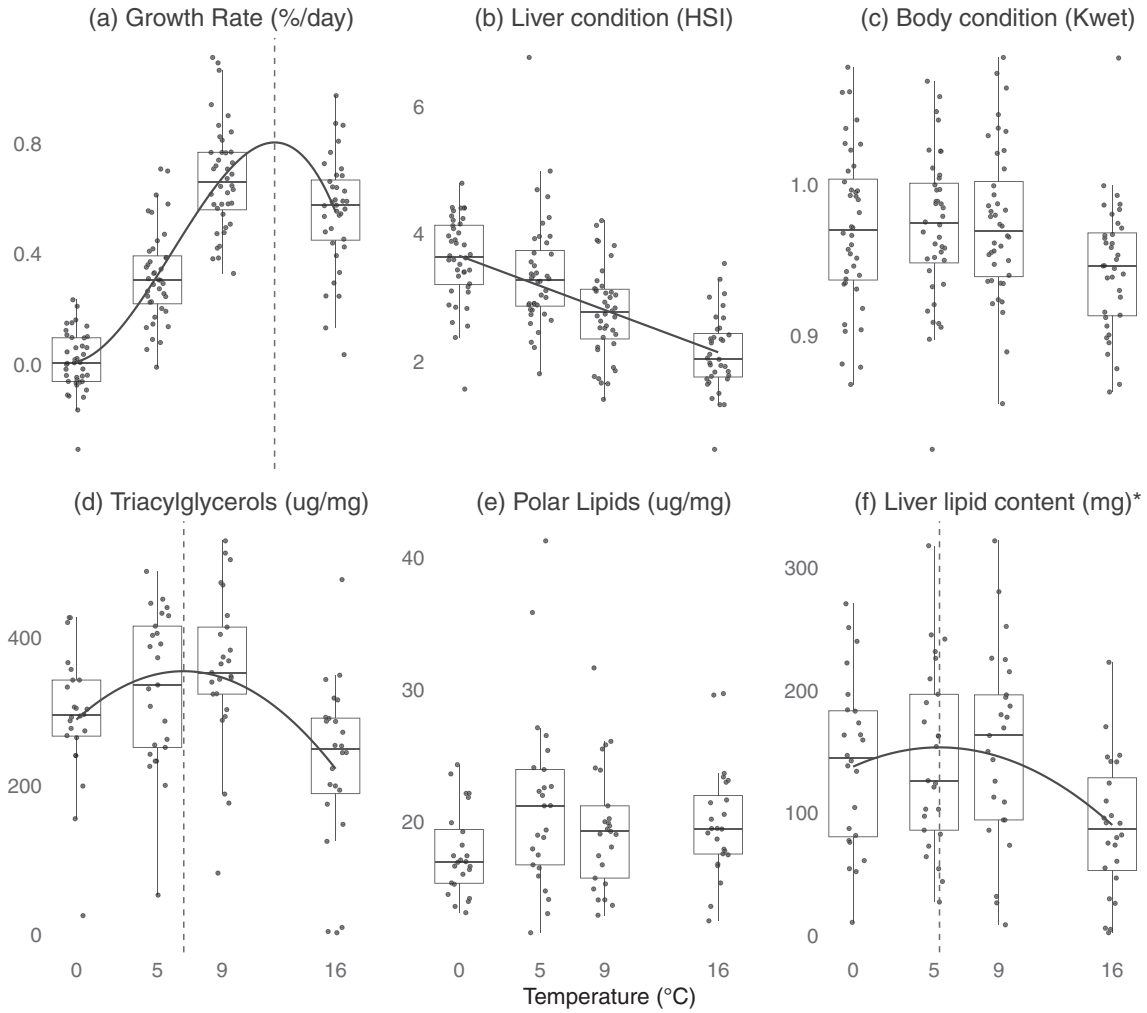


FIGURE 3 | Effects of a 4-week temperature exposure on growth (a), condition (b, c), and liver lipid content (d–f) in juvenile Pacific cod. Growth rate, hepatosomatic index (HSI), and Fulton's condition factor (K_{wet}) were based on wet weight. Models are shown only for metrics significantly affected by temperature (ANOVA). Predicted temperatures of maximal trait values (T_{max}; dashed lines) for non-linear models were 12.2°C for growth rate, 6.6°C for triacylglycerols, and 5.4°C for whole liver lipid content, which was *estimated from the total lipid mass fraction (triacylglycerols, free fatty acids, sterols, and polar lipids) and liver wet weight. Notably, liver lipid content did not differ among the 0°C, 5°C, and 9°C treatments, but was approximately 40% lower on average in the 16°C treatment. K_{wet} and polar lipid mass fraction did not vary significantly among temperature treatments. Free fatty acids (T_{max} = 6.5°C) and sterols (T_{max} = 4.5°C) were also measured but are not shown due to their small contributions to total liver lipid content (1%–2%).

Total lipid content in the liver (mg), which was estimated from the liver size and total lipid mass fraction in the liver, differed significantly among temperature treatments ($F_{3,93} = 4.20$, $p < 0.008$) and was estimated to peak at 5.4°C (Figure 3, Table 2). No differences were observed among individuals at 0°C, 5°C, and 9°C, while those at 16°C had, on average, 36%-42% less total lipid content in the liver compared to the other treatments (Figure 3, Table 2).

3.3 | Correlations Among Growth and Lipid Traits

Liver lipid traits were correlated with each other, such that individuals with larger livers relative to body size (higher HSI) had higher mass fractions of triacylglycerols (TAG; $r = 0.68$, $p < 0.001$) and free fatty acids (FFA; $r = 0.58$, $p < 0.001$), and lower mass fractions of membrane-associated lipids (polar lipids: $r = -0.47$, $p < 0.001$; sterols: $r = -0.23$, $p = 0.04$). These relationships were consistent both across all temperatures and within individual temperature treatments (Figure S5).

Relationships among growth rates and liver lipid traits varied with temperature. There was no significant relationship between growth rates and TAG or FFA mass fraction at the two lowest temperatures (0°C and 5°C), whereas positive correlations were observed at the two highest temperatures (TAG: 9°C: $r = 0.57$, $p = 0.0086$; 16°C: $r = 0.53$, $p = 0.025$).

Notably, liver condition (HSI) and growth rate showed opposite patterns depending on the level of analysis: a negative correlation between HSI and growth rates was observed across all temperatures (SGR_{WW}: $r = -0.26$, $p = 0.003$; SGR_{SL}: $r = -0.25$, $p = 0.004$), whereas positive correlations were observed within individual temperature treatments (e.g., $r = 0.33$–0.45; Figure 4). Correlations among additional lipid classes and traits are provided in Figures S4–S8.

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TABLE 2 | Mean growth rates, condition indices, and liver lipid content by temperature, and parameters of temperature-dependent growth models for metrics significantly affected by temperature.

Mean ± SD				
	0°C	**5°C**	**9°C**	**16°C**
SGR_{SL} (% day^{-1})	0.09 ± 0.04	0.21 ± 0.06	0.31 ± 0.07	0.28 ± 0.08
G_{SL} (mm day^{-1})	0.10 ± 0.05	0.21 ± 0.06	0.32 ± 0.08	0.30 ± 0.09
SGR_{WWT} (% day^{-1})	0.01 ± 0.11	0.32 ± 0.17	0.67 ± 0.19	0.55 ± 0.20
G_{WW} (mg day^{-1})	1.2 ± 12.2	33.6 ± 17.6	78.1 ± 28.4	70.3 ± 30.6
Body condition index (K_{wet})	0.97 ± 0.05	0.97 ± 0.05	0.97 ± 0.05	0.94 ± 0.04
Liver condition index (HSI)	3.61 ± 0.67	3.29 ± 0.69	2.77 ± 0.70	2.13 ± 0.60
Total Liver Lipids (ug·mg^{-1})	317.6 ± 87.8	366.8 ± 101.6	385.2 ± 104.7	255.3 ± 112.3
Liver Triacylgycerols (ug·mg^{-1})	293.4 ± 87.1	336.4 ± 103.0	357.4 ± 105.0	229.7 ± 113.4
Liver Free Fatty Acids (ug·mg^{-1})	4.37 ± 2.05	5.26 ± 2.43	5.55 ± 2.62	2.78 ± 1.58
Liver Sterols (ug·mg^{-1})	2.23 ± 0.47	3.35 ± 0.93	2.78 ± 0.70	2.87 ± 0.54
Liver Polar Lipids (ug·mg^{-1})	17.6 ± 3.2	21.6 ± 6.5	19.4 ± 4.6	19.9 ± 4.2
Est. lipid in whole liver (mg)	140 ± 69	144 ± 76	154 ± 79	89 ± 56
Models and parameters				
	y_o mean ± 1 SE	**a mean ± 1 SE**	**b mean ± 1 SE**	**c mean ± 1 SE**
Cubic model estimates, $Y = y_o + aT + bT^2 + cT^3$				
SGR_{SL} $R^2 = 0.63$	0.093 ± 0.010	0.012 ± 0.008	0.0031 ± 0.0015	−0.00020 ± 0.00006
SGR_{WW} $R^2 = 0.69$	0.011 ± 0.027	0.013 ± 0.022	0.0137 ± 0.0040	−0.00078 ± 0.00017
Liver Sterols $R^2 = 0.24$	2.22 ± 0.14	−0.44 ± 0.32	1.8 ± 0.33	−0.26 ± 0.19
Quadratic model estimates, $Y = y_o + aT + bT^2$				
Liver Total Lipids $R^2 = 0.18$	314.17 ± 20.24	20.21 ± 5.96	−1.49 ± 0.35	—
Liver Triacylgycerols $R^2 = 0.19$	289.20 ± 20.79	19.69 ± 6.11	−1.49 ± 0.36	—
Liver Free Fatty Acids $R^2 = 0.18$	4.30 ± 0.44	0.39 ± 0.13	−0.030 ± 0.0075	—
Est. lipid in whole liver (mg)	137.71 ± 14.0	5.90 ± 4.11	−0.55 ± 0.24	
Linear model estimates, $Y = y_o + aT$				
Hepatosomatic Index $R^2 = 0.40$	3.66 ± 0.09	−0.095 ± 0.0093	—	—

Note: Growth rates are reported as specific growth rate (SGR) and absolute growth rate (G), and modeled using SGR. Body condition (Fulton's condition factor, K_{wet}) and liver condition (hepatosomatic index, HSI) are indices and therefore unitless. Liver lipid values are expressed as mass fractions per wet weight. Estimated whole-liver lipid content was calculated from liver wet weight and total lipid mass fraction (sum of triacylglycerols, free fatty acids, sterols, and polar lipids).

3.4 | Relationships Among Traits and Genetics

Redundancy analysis (RDA), a multivariate method that combines regression and ordination, was used to evaluate relationships among physiological traits and genetic variation using individuals exposed to the highest temperature (16°C). Backward stepwise selection resulted in a final model with three genetic PCs (PC4, PC5, and PC7), explaining 24.5% of total phenotypic variance (adjusted $R^2 = 13.8\%$; $F_{3,21} = 2.28$, $p = 0.031$), 88% of which was captured by the first constrained axis (RDA1). Positive loadings along RDA1 were observed for hepatosomatic index (HSI; 0.72), triacylglycerols (0.69), fatty acids (0.73), total liver lipid content (0.66), and specific growth rate (0.26), whereas polar lipids (−0.69) and sterols (−0.53) had negative loadings. Among the explanatory variables, PC5 had the highest loading on RDA1 (−0.76), consistent with the significant correlations between PC5 scores and liver lipid traits (Figure 5, Table S2).

Functional enrichment analysis of SNPs with the highest loadings on PC5 indicated overrepresentation of genes involved in cell surface signaling and membrane-associated processes. Enriched biological processes included cell adhesion, cytoskeletal organization, calcium signaling, and intracellular signal transduction (Table S3). Many enriched domains (e.g., immunoglobulin-like, cadherin, and fibronectin) and KEGG pathways (e.g., mTOR signaling, oxytocin signaling, platelet activation) are characteristic of membrane or extracellular proteins involved in cell–cell communication, and innate immune function (Figure S9). Together,

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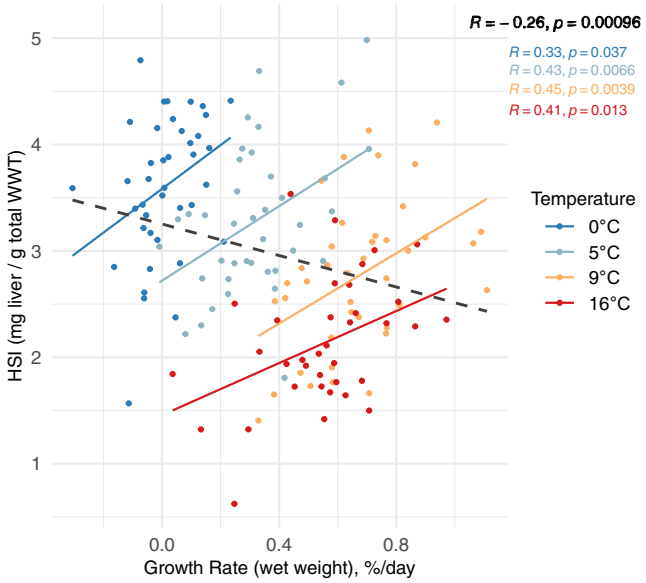


FIGURE 4 | A negative correlation between liver condition (HSI, y-axis) and growth rate (SGR_{WW}, x-axis) was observed across all temperatures (black dashed line), whereas positive correlations were observed within individual temperature treatments.

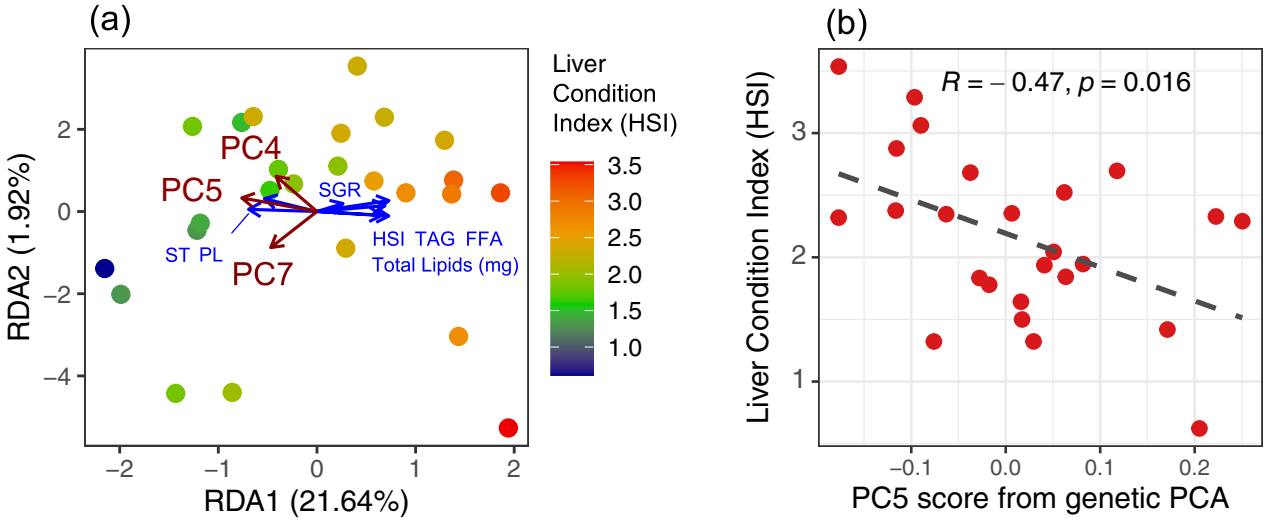


FIGURE 5 | Redundancy analysis (RDA) biplot (a) for juvenile Pacific cod exposed to 16°C, illustrating relationships among liver lipid traits, growth, and genetic principal components (PCs) that explain significant phenotypic variation (PC4, PC5, and PC7). Points are colored by hepatosomatic index (HSI). Principal component 5 (PC5), derived from genotype likelihoods across polymorphic sites in experimental juveniles ($n = 1.6$ M), explained 1.26% of genome-wide genetic variation and had the strongest loading on the primary constrained axis (RDA1), which accounted for 21.64% of phenotypic variation. Consistent with this pattern, PC5 scores were significantly correlated with HSI (b).

these results suggest that PC5 largely captures genetic variation affecting cell communication, immune signaling, and intracellular signaling pathways. Enrichment and gene results for PC4 and PC7 are provided in Appendix S1.

4 | Discussion

Ocean warming has amplified the need to understand factors shaping resilience of ecologically and commercially important species like Pacific cod (Barbeaux et al. 2020). Recent periods of elevated ocean temperatures have been associated with low recruitment rates, likely due in part to reduced survival from spawning through the first summer (Hurst et al. 2010; Laurel et al. 2008; Laurel et al. 2016) and declines in critical liver lipid stores that support overwinter survival (Biro et al. 2005; Laurel, Hicks, et al. 2026). These impacts highlight the need for empirical data on how temperature mediates allocation trade-offs between growth and energy storage, two key fitness traits in juvenile fish.

Here, high-resolution experimental data demonstrate that energy allocation in juvenile Pacific cod is highly temperature-dependent. We found that energy trade-offs are evident only at high temperature, where relatively high growth rates are maintained at the expense of lipid stores. Notably, a 7°C increase from near-optimal conditions (9°C) to 16°C resulted in smaller, less lipid-dense livers with, on average, 42% less total lipid storage compared with much smaller reductions in growth

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rate (~10% based on length and ~18% based on weight). However, within temperatures, growth and lipid allocation varied among individuals and were positively correlated, such that the fastest-growing individuals also accumulated the highest lipid stores, even in the high-temperature treatment. Such variability in fitness-related traits and the genetic mechanisms underlying them may be critical for shaping the adaptive capacity of Pacific cod populations as they encounter temperatures approaching their upper thermal limits.

4.1 | Energetic Tradeoffs Are Only Observed at High Temperature, and Vary Among Individuals

Across gadid species there is a tradeoff between lipid storage and whole-body size at the juvenile stage that corresponds to differences in latitudinal distribution and thermal experience. In Arctic gadids, age-0 juveniles tend to be smaller but more lipid-dense compared to boreal gadids (Copeman et al. 2017). Similar within-species tradeoffs have also been observed among populations of saffron cod, where age-0 juveniles are smaller and more lipid-dense in populations from the Chukchi Sea compared to more southerly Bering Sea populations (Copeman et al. 2016). Copeman et al. (2017) suggested that this tradeoff is related to harsher winters in the Arctic favoring lipid stores and higher predation risk in subarctic regions favoring faster growth, in addition to regional differences such as prey lipid density and constraints on body size due to shorter growing seasons.

Our results support the general hypothesis that growth rate is prioritized in gadids from lower latitudes, and provide new insights into the trade-off between growth and energy storage in Pacific cod. Growth rates were maintained at relatively high levels at the highest temperature at the expense of liver lipids. However, we also found that at 9°C and below, despite major differences in growth rates which impacted HSI, the total amount of lipid stored in the liver did not vary significantly among temperatures (Figure 3f). Thus, there does not appear to be a programmed switch to liver lipid accumulation as temperatures drop and growth rates slow, at least in our experimental fish (e.g., collected post-heatwave and maintained under food replete conditions). In stark contrast, approximately 40% fewer lipids were stored in the liver in individuals exposed to 16°C, indicating a threshold above which high temperatures begin to tilt the balance away from maintaining liver lipid stores to using them.

Additionally, we observed an important distinction in energy allocation responses to temperature at a population and individual level. At the population level, there was a negative correlation between liver condition (HSI) and growth rate (SGR_{WW}) across all temperatures (black dashed line, Figure 4), whereas there were positive correlations between growth and condition for individual fish within temperature treatments (solid lines, Figure 4). Some individuals are therefore able to simultaneously grow faster and store more energy. This challenges the assumption of a strict trade-off between growth and energy storage within individuals, and instead points to individual variation in energetic efficiency or acquisition. These findings support and extend those of experimental work by Copeman et al. (2017), who demonstrated that lipid accumulation rate scales positively with growth rate across a wide thermal range in gadid species. In contrast to that study, which quantified combined liver and muscle lipids, we focused on liver tissue because juvenile cod liver lipid mass fraction is highly responsive to temperature, whereas muscle lipid mass fraction varies comparatively little under similar thermal treatments (Copeman et al. 2017, 2013, 2008). The pronounced thermal sensitivity of liver lipid stores highlights their role as a dynamic energy reserve, suggesting that individuals capable of simultaneously maximizing growth and hepatic energy storage may have a selective advantage in seasonal environments where both body size and stored reserves contribute to overwinter survival. The mechanism of this advantage is not clear and could be due to a variety of physiological, behavioral, or environmental factors (Meager et al. 2018; Goodrich and Clark 2023).

Feeding ration was kept constant across temperature treatments but changes in food availability alongside temperature are an important consideration. Juveniles were well fed and in relatively good condition when they were introduced to the experiment. Growth and condition would likely have differed, particularly in the extreme temperatures, under different prey densities (Sreenivasan and Heintz 2016). Our observations and models may therefore represent best-case scenarios. In natural systems, low or variable prey availability could potentially reduce lipid accumulation even in low temperatures, particularly if these same environmental cues also disrupt the timing, quality, and abundance of prey available. These ecological constraints may therefore amplify differences among individuals, and/or reveal within-individual tradeoffs between somatic growth and liver lipid accumulation not observed in our experiment (Hurst et al. 2018).

4.2 | Larger Juveniles in Heatwaves May Have Greater Capacity for Energy Accumulation

Field surveys have consistently reported larger (but far fewer) age-0 cod during heatwave years linked to earlier spawning timing (Miller et al. 2024), faster growth (Almeida et al. 2024), and selective mortality of smaller individuals in the summer (Thalmann et al. 2024). Our results provide experimental support for the idea that the larger "super survivors" that persisted on nursery grounds in late summer during the 2019 marine heatwave (Thalmann et al. 2024) did not likely reallocate limited energy towards faster growth, but may have been individuals that could acquire and/or assimilate more energy overall. Experimental results also show that fast growing and lipid rich individuals are also more likely to survive winter conditions (Laurel, Hicks, et al. 2026). Collectively, this growing body of work indicates that energy-accumulating traits will likely be favored in warmer years at multiple stages during the first year of life.

4.3 | Genetic Basis For Lipid Allocation Under Thermal Stress

Many of the above findings suggest that major marine heatwave events could act as a selective force during the first year of life, potentially altering the genetic composition of GOA Pacific cod cohorts. This hypothesis depends on fitness-related traits being at least partially influenced by genetics. We generated low-coverage whole-genome sequencing data for all experimental

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individuals (lcWGS), originally to determine their population of origin (Figure 2). However, these data also enabled a preliminary genotype–phenotype analysis to explore whether genetic variation was associated with phenotypic differences in response to the high temperature treatment (16°C). Using redundancy analysis (RDA), we found that ~25% of variation in growth and liver lipid content at 16°C was explained by genetic variation captured by three PC axes (PC4, PC5, and PC7). Of these, PC5 accounted for most of the explained variation (21.6%) and was significantly correlated with liver lipid traits (Figure 5b). Given the complexity of the genetic data (~1.6 million SNPs derived from low-coverage whole-genome sequencing), the amount of phenotypic variation explained by this linear representation of genetic variation (through PCA) is notable, particularly because PC4, PC5, and PC7 each explained only ~1.3% of total genetic variation in our juveniles (~4% total, Figure S2). However, PCs explaining modest proportions of total genetic variance can nonetheless capture biologically meaningful genotype–phenotype relationships, especially when genetic effects influence specific physiological traits rather than overall phenotypic covariance (Aschard et al. 2014).

These results suggest that a portion of variation in lipid storage and energy allocation in juvenile Pacific cod has a genetic basis. The large "super-survivors" observed in nurseries during heatwave years (Thalmann et al. 2024) may therefore reflect selective filtering for individuals with greater capacity for energy acquisition and storage under high temperature. Such genotype–phenotype links are expected for complex fitness traits and are routinely exploited in breeding programs that select for performance-related characteristics (Houston et al. 2020). Interestingly, although growth rate and liver lipid content were positively correlated among individuals within temperature treatments, genetic associations were detected only for lipid-related traits. This pattern may reflect differences in how these traits respond to temperature: liver lipid storage is highly temperature-sensitive and dynamically regulated, whereas growth rate, although reduced at high temperature, was comparatively less responsive. Lipid allocation may therefore represent a trait with greater detectable genetic variability or stronger potential for selection under thermal stress. It is also possible that genetic influences on growth are distributed across additional genomic axes not captured in the first 10 PCs examined here. Nonetheless, the absence of shared genetic signals along the same PC axes suggests that growth and lipid allocation are not necessarily controlled by tightly linked genetic regions.

Importantly, most phenotypic variation (~75%) remained unexplained by the genetic PCs, suggesting that additional factors such as developmental or transgenerational carryover effects and epigenetic processes may also play substantial roles in shaping juvenile responses to thermal stress. (Hurst et al. 2012; Leung and McAfee 2020; Marshall and Connallon 2023; Eirin-Lopez and Putnam 2019; Houston et al. 2020).

4.4 | Post-Heatwave Era Juveniles May Have Higher Thermal Optima

Our findings raise the question of whether selection for enhanced growth and energy assimilation is ongoing in the GOA, and whether changes coincide with shifts in thermal optima. While growth followed the typical asymmetric thermal performance curve, the thermal optimum (T_{max}) for growth in this experiment was 12.2°C, 0.7°C higher than a previously reported T_{max} of 11.5°C for juveniles collected a decade earlier (Laurel, Spencer et al. 2016). The juveniles in our study were collected in summer 2022, a non-heatwave year, but may have descended from cohorts that experienced major marine heatwaves occurring 3–8 years earlier in the GOA, between 2014 and 2016 and in 2019. Other modulators of thermal performance, such as acclimation temperature, feeding ration, and body size, were either matched or differed in ways expected to lower T_{max}, not increase it. For example, juveniles in this study were larger than in the previous study (Laurel, Spencer et al. 2016), and larger body size is associated with lower T_{max} (Imsland et al. 2005; Schakmann et al. 2023). It would be interesting to compare genetics of contemporary GOA juveniles to historic populations (from archived tissue), or assess genetics by cohort in adults, particularly at variants and genes we found to be associated with lipid content in our study, to determine whether there is evidence of selection occurring during MHWs. Of course, other non-genetic factors not easily accounted for could cause thermal optima to differ among cohorts (environmental history of the individual, parents via physiological remodeling or epigenetic factors) (Hurst et al. 2012; Leung and McAfee 2020; Marshall and Connallon 2023; Eirin-Lopez and Putnam 2019).

Evidence from salmonids supports both inherited and developmental effects on thermal preference. Juvenile Atlantic salmon exhibit thermal optima aligned with their source population (Jonsson 2023), and high incubation temperatures can cause lasting increases in feeding and growth rates that persist for years post-hatch (Finstad and Jonsson 2012; Jonsson 2023). Similarly, Pacific cod collected during warm years have been shown to grow faster at elevated temperatures than those from colder years (Hurst et al. 2012), suggesting that thermal history can influence temperature preference or performance. Our elevated thermal optimum estimates may therefore reflect recent selection or plastic developmental responses in a population shaped by historical thermal events.

5 | Conclusion

Low abundances of Pacific cod in the Gulf of Alaska coincided with a series of marine heatwaves in the late 2010s, prompting fishery closures and raising concerns about the species' vulnerability to rising ocean temperatures (Laurel et al. 2023). Prolonged thermal stress likely resulted in cumulative mortality from hatch through the first winter (Laurel, Hicks, et al. 2026). However, the apparent selective mortality of smaller individuals, and retention of larger juveniles, in summer during these events (Thalmann et al. 2024), combined with our experimental finding that larger individuals also possess greater lipid reserves, suggests that a subset of individuals may harbor genetically defined traits that improve survival outcomes. Our functional analysis of trait-associated genetic variation points to genes involved in cellular signaling and membrane dynamics (Figure S9), processes that may influence thermal acclimation (e.g., membrane fluidity and signaling pathways). However, it remains unclear whether this genetic variation is causal or

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instead reflects linkage with other genes that, for example, enhance energy acquisition (e.g., foraging efficiency) or assimilation (e.g., metabolic efficiency).

As higher-resolution genomic datasets become available, it may be possible to identify specific genotypes associated with resilience, or vulnerability, to thermal stress, enabling population monitoring for signals of selective sweeps and adaptive change. Such genomic indicators, combined with temperature-dependent models of growth and liver lipid allocation, could be used to improve recruitment forecasting during future marine heatwaves. In addition, identifying climate-resilient genetic groups may inform future management decisions, particularly as stock boundaries, connectivity, and population structure in the Gulf of Alaska and eastern Bering Sea are reevaluated.

Author Contributions

Laura H. Spencer: investigation, writing – original draft, methodology, validation, writing – review and editing, visualization, formal analysis, project adminiatration, conceptualization. **Mary Beth Rew Hicks:** investigation, writing – original draft, writing – review and editing, methodology, resources, data curation. **Ingrid B. Spies:** conceptualization, funding acquisition, writing – review and editing, resources, supervision, project administration, methodology, investigation. **Benjamin J. Laurel:** conceptualization, investigation, funding acquisition, writing – review and editing, validation, methodology, supervision, project administration, resources, data curation. **Louise A. Copeman:** conceptualization, investigation, writing – original draft, writing – review and editing, methodology, project administration, resources, supervision, data curation. **Steven B. Roberts:** project administration, supervision, resources, funding acquisition, conceptualization.

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Ethics Statement

This research was carried out in accordance with all applicable institutional and national guidelines. Pacific cod embryos were collected under an Alaska Department of Fish and Game Permit #CF-22-097. This research was conducted at NOAA's Alaska Fisheries Science Center Laboratory in Newport, Oregon. NOAA National Marine Fisheries Service does not have an Institutional Animal Care and Use Committee (IACUC) approval process for research on fishes. All work followed American Fisheries Society policies on the Guidelines for Use of Fishes in Research (American Fisheries Society: https://fisheries.org/docs/policy_useoffishes.pdf) and the AVMA Guidelines on Euthanasia (American Veterinary Medical Association: https://olaw.nih.gov/sites/default/files/Euthanasia2007.pdf).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Raw sequence data is publicly available on the NCBI Sequence Read Archive (SRA) under BioProject PRJNA1298946. Phenotype data is available on the GitHub repository, https://github.com/RobertsLab/project-cod-temperature.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** PCA biplots (PC1xPC2) constructed from a covariance matrix using all SNPs in juvenile only (n = ~1.6 M), with point colors showing temperature treatment (top), and mortality in 16C treatment (bottom). PC1 is likely associated with population structure along the wGOA/eGOA gradient. PC2 is not associated with population, but tends to be associated with mortality in 16C treatment. **Figure S2:** Scree-plot of PCs 1–50, from PCA with polymorphic sites (SNPs) experimental juveniles (n = ~1.6 M), with PC axes 1–10 explaining a cumulative total of 12.1% genetic variation. **Figure S3:** PC axes that represent the wGOA/eGOA structure/gradient using juveniles + reference adults (y-axis, PC1 from Figure 2), and only juveniles (x-axis). PCAs were constructed from covariance matrix using all SNPs in juvenile only (n = ~1.6 M) or from sites intersecting with the reference adult dataset (n = ~285 K). **Figure S4:** (a) Pearson correlation coefficients among condition indices, growth rates, and lipid components, calculated across all temperatures. (b) Pearson correlations among liver condition index (HSI) and growth and liver traits, highlighting consistent positive relationships between HSI and liver lipid content (green) and triacylglycerols, the dominant lipid component, across temperatures. In contrast, correlations involving K_{wet} were inconsistent (blue), and the relationship between HSI and growth rate (wet weight) differed when comparing across all temperatures versus within temperatures (red). **Figure S5:** Pearson correlation coefficients among liver and growth/size metrics. Data include individuals within each temperature treatment. **Figure S6:** Pearson correlations among SGR_{ww} and other traits across temperatures. **Figure S7:** Pearson correlations among K_{wet} and other traits across temperatures. **Figure S8:** Pearson correlations among TAG concentration in the liver and other traits across temperatures. **Figure S9:** Top enriched biological processes (Uniprot Keywords, GO terms) and KEGG pathways for genes containing SNPs with the highest PC loadings, shown for PCs that significantly explained phenotype variation in the highest temperature (16C) according to RDA (PC4, PC5, PC7). **Table S1:** Results of redundancy analysis (RDA) examining associations between genetic variation and physiological traits in juvenile Pacific cod exposed to 16°C. The final model included genetic PCs 4, 5, and 7 as explanatory variables. Eigenvalues, proportion of variance explained, loadings of physiological traits, and genetic PC axes on the constrained ordination axes are shown. **Table S2:** Correlations among genetic PCs and traits in RDA model, with correlations meeting the FDR < 0.1 italicized. **Table S3:** Genes containing 10+ top SNPs contributing to PC scores, based on PC loadings.

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